Class 09

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Background

In today's mini project we will analyse candy data with the exploratory graphics, basic statistics, correlation analysis and principal component analysis methods we have been learning thus far.

We are using FiveThirtyEight's Halloween Candy dataset. Your task is to explore their candy dataset to find out answers to these types of questions: What is the top ranked snack-sized Halloween candy? What made some candies more desirable than others? Was it price? Maybe it was just sugar content?

Data Import

The data comes as a CSV file from 538.

```
read.csv("candy-data.csv")
```

	competitorname	chocolate	fruity	caramel	peanutyalmondy	nougat
1	100 Grand	1	0	1	0	0
2	3 Musketeers	1	0	0	0	1
3	One dime	0	0	0	0	0
4	One quarter	0	0	0	0	0
5	Air Heads	0	1	0	0	0
6	Almond Joy	1	0	0	1	0
7	Baby Ruth	1	0	1	1	1
8	Boston Baked Beans	0	0	0	1	0
9	Candy Corn	0	0	0	0	0
10	Caramel Apple Pops	0	1	1	0	0
11	Charleston Chew	1	0	0	0	1
12	Chewey Lemonhead Fruit Mix	0	1	0	0	0
13	Chiclets	0	1	0	0	0
14	Dots	0	1	0	0	0
15	Dum Dums	0	1	0	0	0
16	Fruit Chews	0	1	0	0	0
17	Fun Dip	0	1	0	0	0
18	Gobstopper	0	1	0	0	0
19	Haribo Gold Bears	0	1	0	0	0
20	Haribo Happy Cola	0	0	0	0	0
21	Haribo Sour Bears	0	1	0	0	0
22	Haribo Twin Snakes	0	1	0	0	0
23	Hershey's Kisses	1	0	0	0	0
24	Hershey's Krackel	1	0	0	0	0
25	Hershey's Milk Chocolate	1	0	0	0	0
26	Hershey's Special Dark	1	0	0	0	0
27	Jawbusters	0	1	0	0	0
28	Junior Mints	1	0	0	0	0
29	Kit Kat	1	0	0	0	0
30	Laffy Taffy	0	1	0	0	0
31	Lemonhead	0	1	0	0	0
32	Lifesavers big ring gummies	0	1	0	0	0
33	Peanut butter M&M's	1	0	0	1	0
34	M&M's	1	0	0	0	0
35	Mike & Ike	0	1	0	0	0
36	Milk Duds	1	0	1	0	0
37	Milky Way	1	0	1	0	1
38	Milky Way Midnight	1	0	1	0	1
39	Milky Way Simply Caramel	1	0	1	0	0
40	Mounds	1	0	0	0	0
41	Mr Good Bar	1	0	0	1	0
42	Nerds	0	1	0	0	0

43	Nestle Butterfinger	1	0	0	1	0
44	Nestle Crunch	1	0	0	0	0
45	Nik L Nip	0	1	0	0	0
46	Now & Later	0	1	0	0	0
47	Payday	0	0	0	1	1
48	Peanut M&Ms	1	0	0	1	0
49	Pixie Sticks	0	0	0	0	0
50	Pop Rocks	0	1	0	0	0
51	Red vines	0	1	0	0	0
52	Reese's Miniatures	1	0	0	1	0
53	Reese's Peanut Butter cup	1	0	0	1	0
54	Reese's pieces	1	0	0	1	0
55	Reese's stuffed with pieces	1	0	0	1	0
56	Ring pop	0	1	0	0	0
57	Rolo	1	0	1	0	0
58	Root Beer Barrels	0	0	0	0	0
59	Runts	0	1	0	0	0
60	Sixlets	1	0	0	0	0
61	Skittles original	0	1	0	0	0
62	Skittles wildberry	0	1	0	0	0
63	Nestle Smarties	1	0	0	0	0
64	Smarties candy	0	1	0	0	0
65	Snickers	1	0	1	1	1
66	Snickers Crisper	1	0	1	1	0
67	Sour Patch Kids	0	1	0	0	0
68	Sour Patch Tricksters	0	1	0	0	0
69	Starburst	0	1	0	0	0
70	Strawberry bon bons	0	1	0	0	0
71	Sugar Babies	0	0	1	0	0
72	Sugar Daddy	0	0	1	0	0
73	Super Bubble	0	1	0	0	0
74	Swedish Fish	0	1	0	0	0
75	Tootsie Pop	1	1	0	0	0
76	Tootsie Roll Juniors	1	0	0	0	0
77	Tootsie Roll Midgies	1	0	0	0	0
78	Tootsie Roll Snack Bars	1	0	0	0	0
79	Trolli Sour Bites	0	1	0	0	0
80	Twix	1	0	1	0	0
81	Twizzlers	0	1	0	0	0
82	Warheads	0	1	0	0	0
83	Welch's Fruit Snacks	0	1	0	0	0
84	Werther's Original Caramel	0	0	1	0	0
85	Whoppers	1	0	0	0	0

	crispedricewafer	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
1	1	0	1	0	0.732	0.860	66.97173
2	0	0	1	0	0.604	0.511	67.60294
3	0	0	0	0	0.011	0.116	32.26109
4	0	0	0	0	0.011	0.511	46.11650
5	0	0	0	0	0.906	0.511	52.34146
6	0	0	1	0	0.465	0.767	50.34755
7	0	0	1	0	0.604	0.767	56.91455
8	0	0	0	1	0.313	0.511	23.41782
9	0	0	0	1	0.906	0.325	38.01096
10	0	0	0	0	0.604	0.325	34.51768
11	0	0	1	0	0.604	0.511	38.97504
12	0	0	0	1	0.732	0.511	36.01763
13	0	0	0	1	0.046	0.325	24.52499
14	0	0	0	1	0.732	0.511	42.27208
15	0	1	0	0	0.732	0.034	39.46056
16	0	0	0	1	0.127	0.034	43.08892
17	0	1	0	0	0.732	0.325	39.18550
18	0	1	0	1	0.906	0.453	46.78335
19	0	0	0	1	0.465	0.465	57.11974
20	0	0	0	1	0.465	0.465	34.15896
21	0	0	0	1	0.465	0.465	51.41243
22	0	0	0	1	0.465	0.465	42.17877
23	0	0	0	1	0.127	0.093	55.37545
24	1	0	1	0	0.430	0.918	62.28448
25	0	0	1	0	0.430	0.918	56.49050
26	0	0	1	0	0.430	0.918	59.23612
27	0	1	0	1	0.093	0.511	28.12744
28	0	0	0	1	0.197	0.511	57.21925
29	1	0	1	0	0.313	0.511	76.76860
30	0	0	0	0	0.220	0.116	41.38956
31	0	1	0	0	0.046	0.104	39.14106
32	0	0	0	0	0.267	0.279	52.91139
33	0	0	0	1	0.825	0.651	71.46505
34	0	0	0	1	0.825	0.651	66.57458
35	0	0	0	1	0.872	0.325	46.41172
36	0	0	0	1	0.302	0.511	55.06407
37	0	0	1	0	0.604	0.651	73.09956
38	0	0	1	0	0.732	0.441	60.80070
39	0	0	1	0	0.965	0.860	64.35334
40	0	0	1	0	0.313	0.860	47.82975
41	0	0	1	0	0.313	0.918	54.52645
42	0	1	0	1	0.848	0.325	55.35405

43	0	0	1	0	0.604	0.767	70.73564
44	1	0	1	0	0.313	0.767	66.47068
45	0	0	0	1	0.197	0.976	22.44534
46	0	0	0	1	0.220	0.325	39.44680
47	0	0	1	0	0.465	0.767	46.29660
48	0	0	0	1	0.593	0.651	69.48379
49	0	0	0	1	0.093	0.023	37.72234
50	0	1	0	1	0.604	0.837	41.26551
51	0	0	0	1	0.581	0.116	37.34852
52	0	0	0	0	0.034	0.279	81.86626
53	0	0	0	0	0.720	0.651	84.18029
54	0	0	0	1	0.406	0.651	73.43499
55	0	0	0	0	0.988	0.651	72.88790
56	0	1	0	0	0.732	0.965	35.29076
57	0	0	0	1	0.860	0.860	65.71629
58	0	1	0	1	0.732	0.069	29.70369
59	0	1	0	1	0.872	0.279	42.84914
60	0	0	0	1	0.220	0.081	34.72200
61	0	0	0	1	0.941	0.220	63.08514
62	0	0	0	1	0.941	0.220	55.10370
63	0	0	0	1	0.267	0.976	37.88719
64	0	1	0	1	0.267	0.116	45.99583
65	0	0	1	0	0.546	0.651	76.67378
66	1	0	1	0	0.604	0.651	59.52925
67	0	0	0	1	0.069	0.116	59.86400
68	0	0	0	1	0.069	0.116	52.82595
69	0	0	0	1	0.151	0.220	67.03763
70	0	1	0	1	0.569	0.058	34.57899
71	0	0	0	1	0.965	0.767	33.43755
72	0	0	0	0	0.418	0.325	32.23100
73	0	0	0	0	0.162	0.116	27.30386
74	0	0	0	1	0.604	0.755	54.86111
75	0	1	0	0	0.604	0.325	48.98265
76	0	0	0	0	0.313	0.511	43.06890
77	0	0	0	1	0.174	0.011	45.73675
78	0	0	1	0	0.465	0.325	49.65350
79	0	0	0	1	0.313	0.255	47.17323
80	1	0	1	0	0.546	0.906	81.64291
81	0	0	0	0	0.220	0.116	45.46628
82	0	1	0	0	0.093	0.116	39.01190
83	0	0	0	1	0.313	0.313	44.37552
84	0	1	0	0	0.186	0.267	41.90431
85	1	0	0	1	0.872	0.848	49.52411

```r
candy_file <- "candy-data.csv"
candy = read.csv(candy_file, row.names= 1)
head(candy)
```

```
             chocolate fruity caramel peanutyalmondy nougat crispedricewafer
100 Grand            1      0       1              0      0                1
3 Musketeers         1      0       0              0      1                0
One dime             0      0       0              0      0                0
One quarter          0      0       0              0      0                0
Air Heads            0      1       0              0      0                0
Almond Joy           1      0       0              1      0                0
             hard bar pluribus sugarpercent pricepercent winpercent
100 Grand       0   1        0        0.732        0.860   66.97173
3 Musketeers    0   1        0        0.604        0.511   67.60294
One dime        0   0        0        0.011        0.116   32.26109
One quarter     0   0        0        0.011        0.511   46.11650
Air Heads       0   0        0        0.906        0.511   52.34146
Almond Joy      0   1        0        0.465        0.767   50.34755
```

“‘

Q How many different candy types are in this datasubset?

There are 85 rows in this dataset

Q How many fruity candy types are in the dataset?

```r
sum(candy$fruity)
```

```
[1] 38
```

One of the most interesting variables in the dataset is winpercent. For a given candy this value is the percentage of people who prefer this candy over another randomly chosen candy from the dataset (what 538 term a matchup). Higher values indicate a more popular candy.

```r
candy["Twix", ]$winpercent
```

```
[1] 81.64291
```

an alternative way:

```r
library(dplyr)
```

```
Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
     winpercent
Twix   81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is it's winpercent value?

```r
candy["M&M's", ]$winpercent
```

```
[1] 66.57458
```

Q4. What is the winpercent value for "Kit Kat"?

```r
candy["Kit Kat", ]$winpercent
```

```
[1] 76.7686
```

Q5. What is the winpercent value for "Tootsie Roll Snack Bars"?

```r
candy["Tootsie Roll Snack Bars", ]$winpercent
```

```
[1] 49.6535
```

Side-note: the skimr::skim() function

There is a useful `skim()` function in the skimr package that can help give you a quick overview of a given dataset. Let's install this package and try it on our candy data.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyalmondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedricewafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

winpercent seems to be on a different scale, as it goes over 50.

Q7. What do you think a zero and one represent for the candy$chocolate column?

The candy either does or doesnt contain chocolate

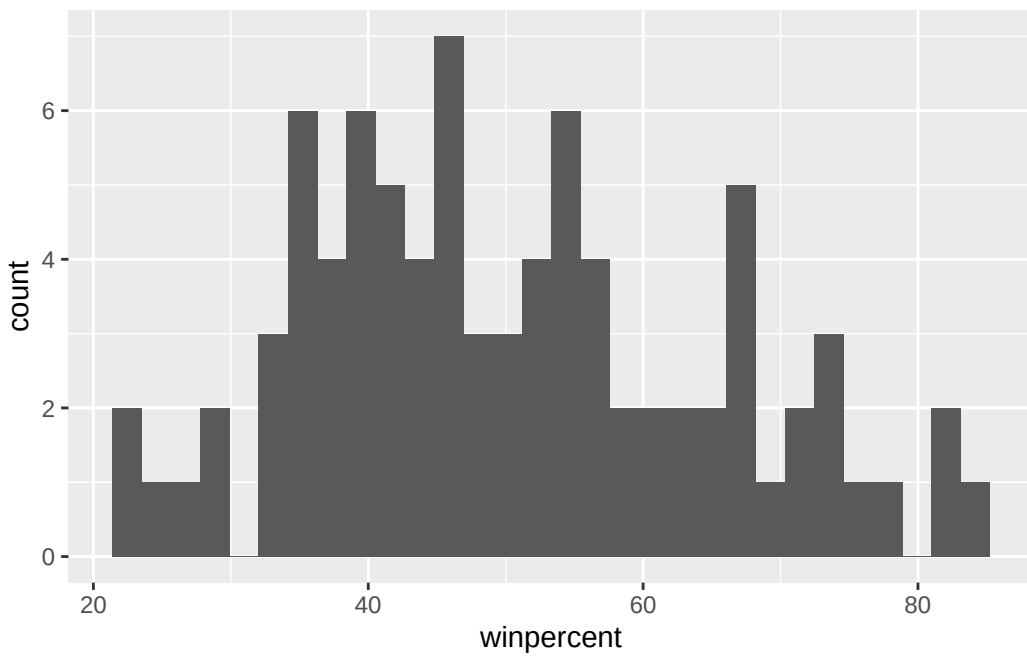
Exploratory Analysis

Q8. Plot a histogram of winpercent values using both base R an ggplot2.

```
library(ggplot2)
```

```
ggplot(candy) + aes(winpercent) + geom_histogram()
```

```
`stat_bin()` using `bins = 30`. Pick better value `binwidth`.
```

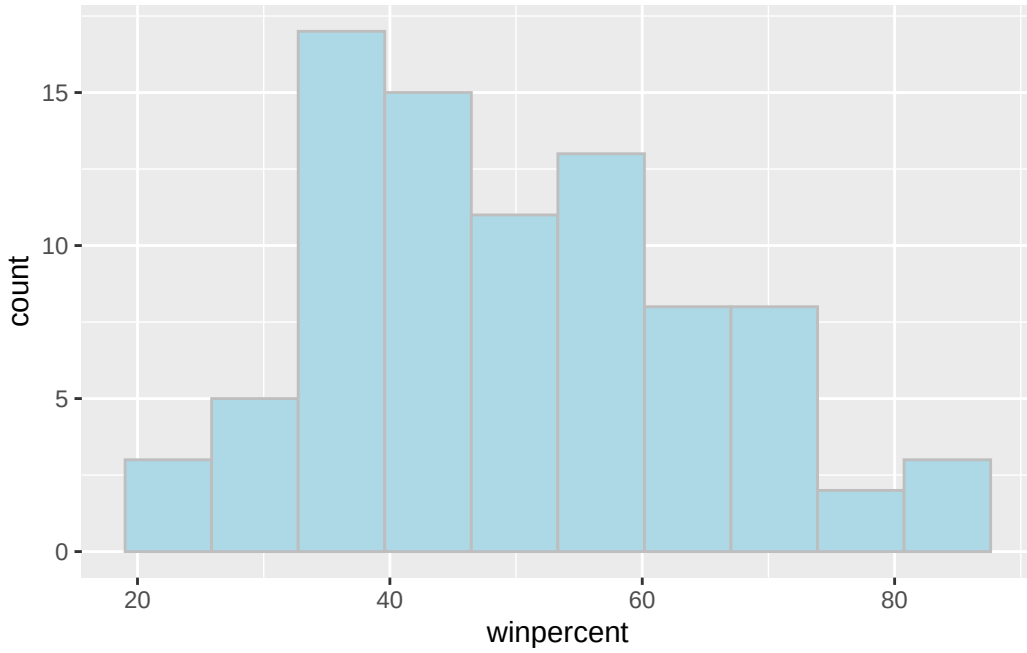


Q9. Is the distribution of winpercent values symmetrical?

No it is not.

Let's make the histogram a bit nicer

```r
ggplot(candy) + aes(winpercent) + geom_histogram(bins=10, fill="lightblue", col="gray")
```



Q10. Is the center of the distribution above or below 50%?

Below- the highest bin is below 40%

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

Steps: 1. Find all chocolate candy in the dataset 2. Extract or find their winpercent values 3. Calculate the mean if these values

4. Find all fruit candy in the dataset
5. Extract or find their winpercent values
6. Calculate the mean of these values

```r
choc.candy <- candy[ candy$chocolate==1, ]
```

```r
choc.win <- choc.candy$winpercent
mean(choc.win)
```

```
[1] 60.92153
```

```r
fruit.candy <- candy[ candy$fruity==1, ]
```

```r
fruit.win <- fruit.candy$winpercent
mean(fruit.win)
```

```
[1] 44.11974
```

So, the mean chocolate is higher

Q12. Is this difference statistically significant?

```r
t.test(choc.win, fruit.win)
```

```
    Welch Two Sample t-test

data:  choc.win and fruit.win
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

Yes as the p value is very small

Overall Candy Rankings

Let's use the base R order() function together with head() to sort the whole dataset by winpercent. Or if you have been getting into the tidyverse and the dplyr package you can use the arrange() function together with head() to do the same thing and answer the following questions:

Q13. What are the five least liked candy types in this set?

```r
y <- c("z", "b", "c")
sort(y)
```

```
[1] "b" "c" "z"
```

gonna rearrange it alphabetically, numerically etc

```r
y
```

```
[1] "z" "b" "c"
```

```r
order(y)
```

```
[1] 2 3 1
```

```r
y[ order(y)]
```

```
[1] "b" "c" "z"
```

```r
sort(candy$winpercent)
```

```
 [1] 22.44534 23.41782 24.52499 27.30386 28.12744 29.70369 32.23100 32.26109
 [9] 33.43755 34.15896 34.51768 34.57899 34.72200 35.29076 36.01763 37.34852
[17] 37.72234 37.88719 38.01096 38.97504 39.01190 39.14106 39.18550 39.44680
[25] 39.46056 41.26551 41.38956 41.90431 42.17877 42.27208 42.84914 43.06890
[33] 43.08892 44.37552 45.46628 45.73675 45.99583 46.11650 46.29660 46.41172
[41] 46.78335 47.17323 47.82975 48.98265 49.52411 49.65350 50.34755 51.41243
[49] 52.34146 52.82595 52.91139 54.52645 54.86111 55.06407 55.10370 55.35405
[57] 55.37545 56.49050 56.91455 57.11974 57.21925 59.23612 59.52925 59.86400
[65] 60.80070 62.28448 63.08514 64.35334 65.71629 66.47068 66.57458 66.97173
[73] 67.03763 67.60294 69.48379 70.73564 71.46505 72.88790 73.09956 73.43499
[81] 76.67378 76.76860 81.64291 81.86626 84.18029
```

```r
inds <- order(candy$winpercent)
head(candy[inds, ], 5)
```

```
                   chocolate fruity caramel peanutyalmondy nougat
Nik L Nip                  0      1       0              0      0
Boston Baked Beans         0      0       0              1      0
Chiclets                   0      1       0              0      0
Super Bubble               0      1       0              0      0
Jawbusters                 0      1       0              0      0
                   crispedricewafer hard bar pluribus sugarpercent pricepercent
Nik L Nip                         0    0   0        1        0.197        0.976
```

```
Boston Baked Beans                0    0   0        1        0.313        0.511
Chiclets                          0    0   0        1        0.046        0.325
Super Bubble                      0    0   0        0        0.162        0.116
Jawbusters                        0    1   0        1        0.093        0.511
                   winpercent
Nik L Nip            22.44534
Boston Baked Beans   23.41782
Chiclets             24.52499
Super Bubble         27.30386
Jawbusters           28.12744
```

head and then comma 5 will show the last five order puts the numerical order but their position in the table.

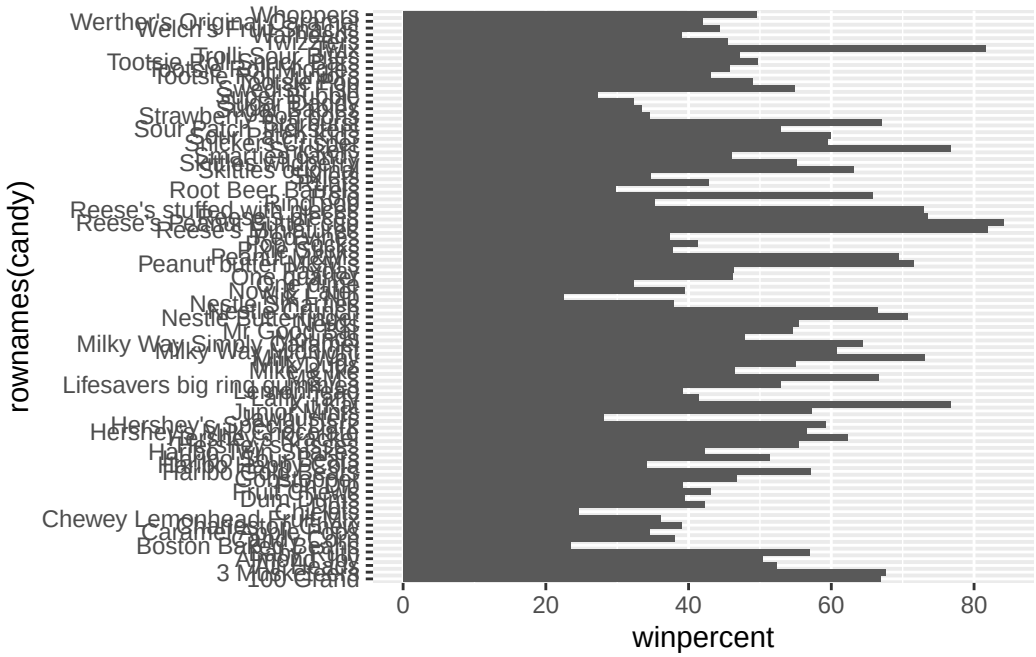
Q14. What are the top 5 all time favorite candy types out of this set?

```r
inds <- order(candy$winpercent)
tail(candy[inds, ], 5)
```

```
                          chocolate fruity caramel peanutyalmondy nougat
Snickers                          1      0       1              1      1
Kit Kat                           1      0       0              0      0
Twix                              1      0       1              0      0
Reese's Miniatures                1      0       0              1      0
Reese's Peanut Butter cup         1      0       0              1      0
                          crispedricewafer hard bar pluribus sugarpercent
Snickers                                 0    0   1        0        0.546
Kit Kat                                  1    0   1        0        0.313
Twix                                     1    0   1        0        0.546
Reese's Miniatures                       0    0   0        0        0.034
Reese's Peanut Butter cup                0    0   0        0        0.720
                          pricepercent winpercent
Snickers                         0.651   76.67378
Kit Kat                          0.511   76.76860
Twix                             0.906   81.64291
Reese's Miniatures               0.279   81.86626
Reese's Peanut Butter cup        0.651   84.18029
```

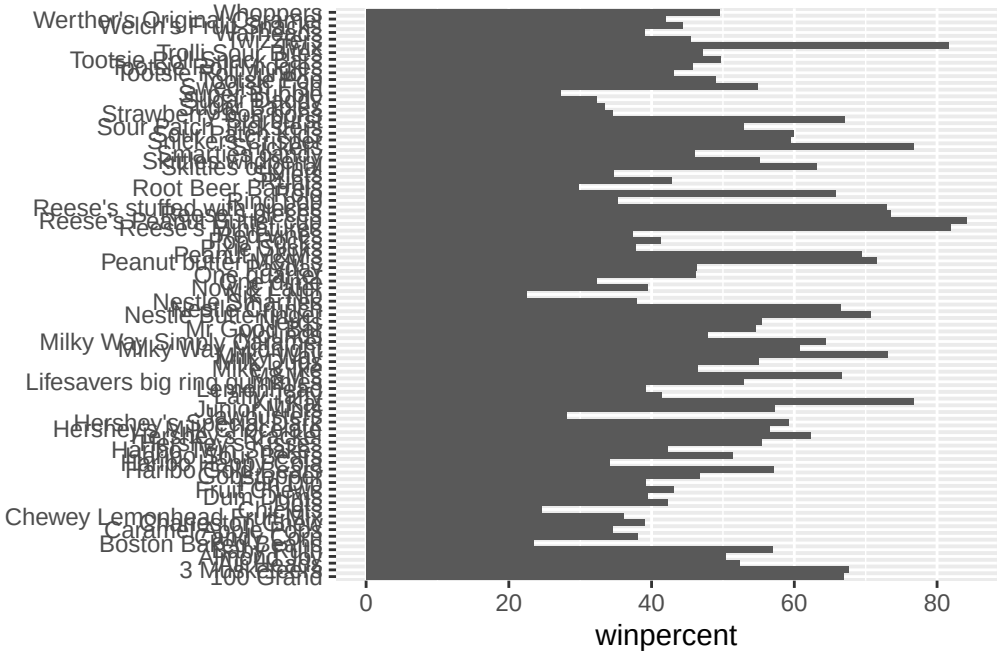
Q15. Make a first barplot of candy ranking based on winpercent values.

```r
ggplot(candy) + aes(winpercent, rownames(candy)) + geom_col()
```



cant read!!!

```
ggplot(candy) + aes(winpercent, rownames(candy)) + geom_col() + ylab("") # turn y label off t
```

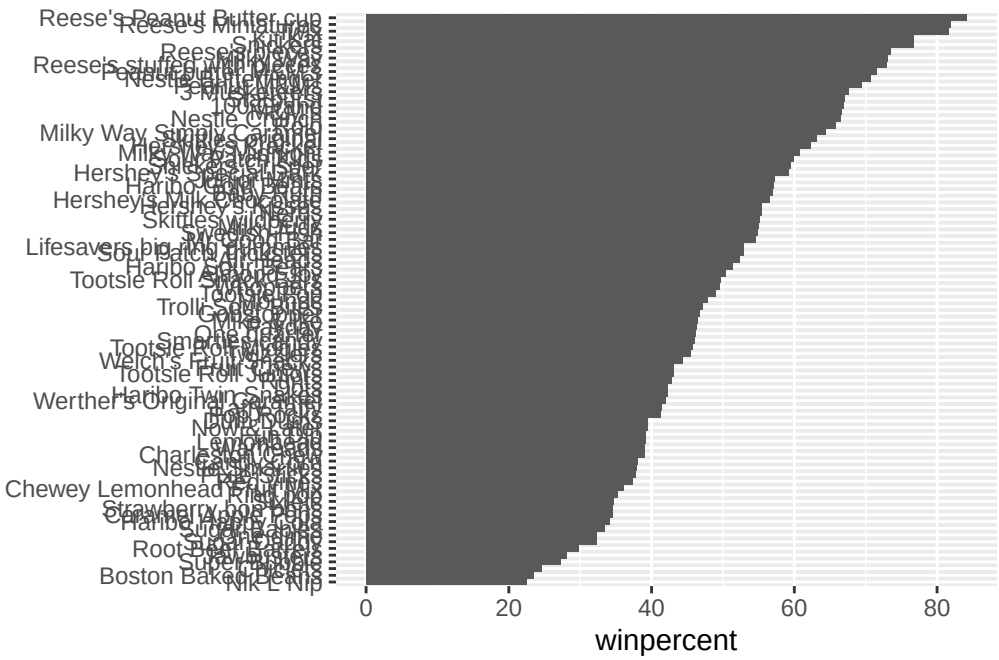


```
ggsave("barplot1.png", height=10, width=4)
```

bit of a fudge, but youll have to play around

> Q16. This is quite ugly, use the reorder() function to get the bars sorted by winpercent?

```
ggplot(candy) + aes(winpercent,
                    reorder( rownames(candy), winpercent)) + geom_col() + ylab("") # turn y l
```




```
ggsave("barplot2.png", height=10, width=4)
```

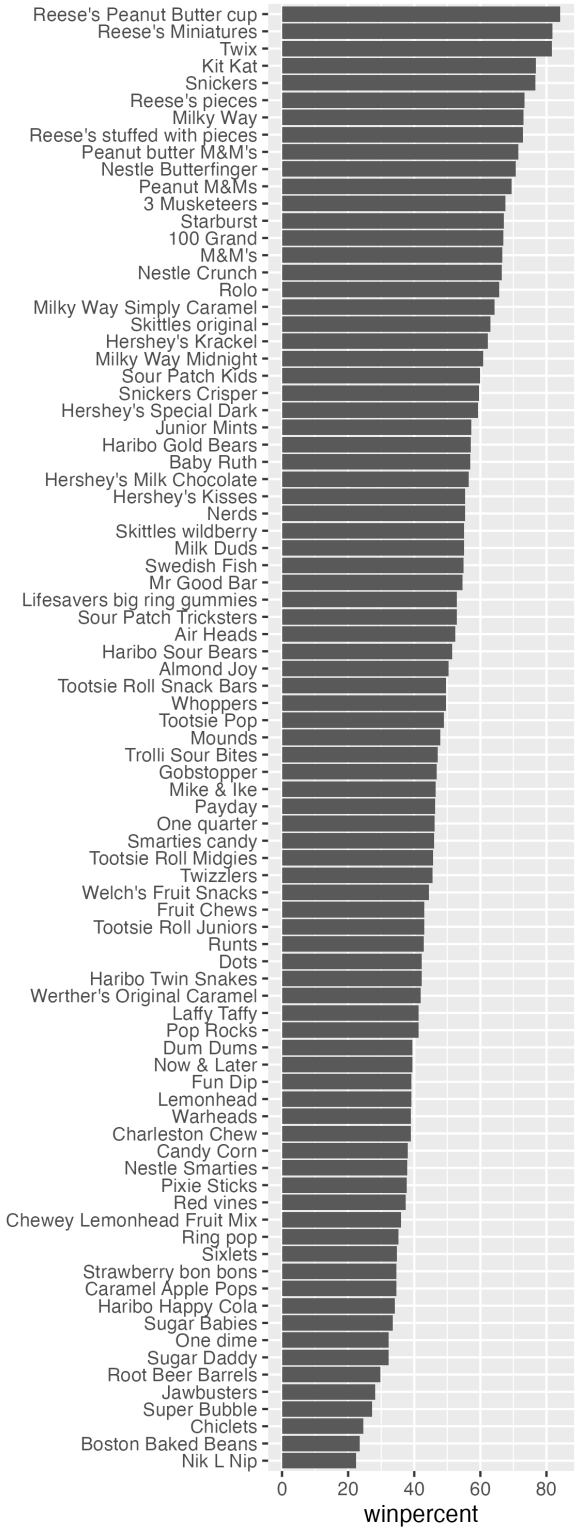
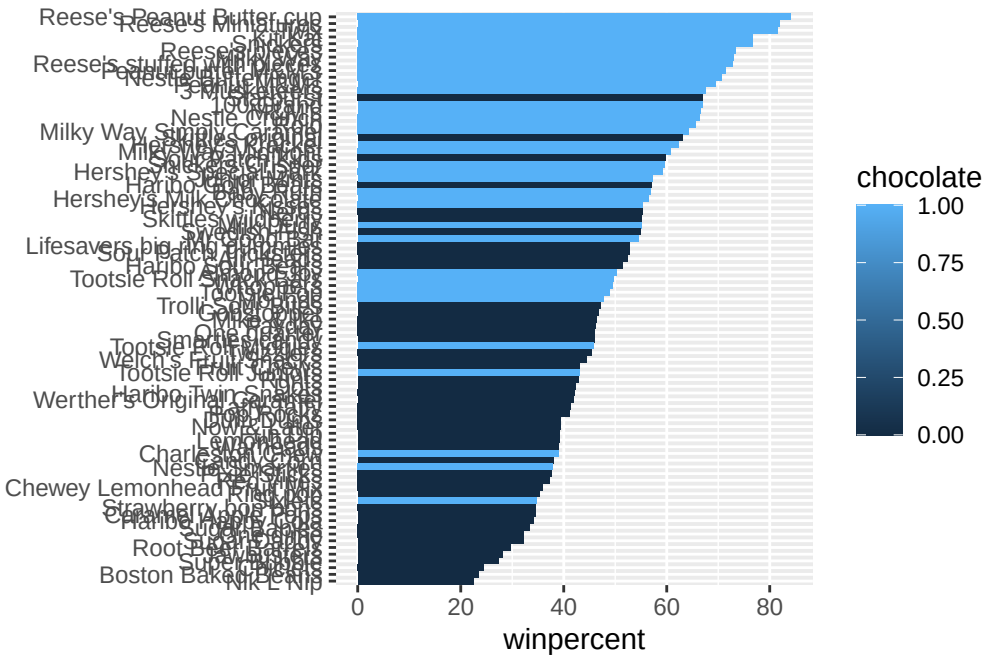


Figure 1: My second barplot for Q16

```r
ggplot(candy) + aes(winpercent,
                    reorder( rownames(candy), winpercent),
                    fill=chocolate ) +
  geom_col() +
  ylab("") # turn y label off that we dont need
```



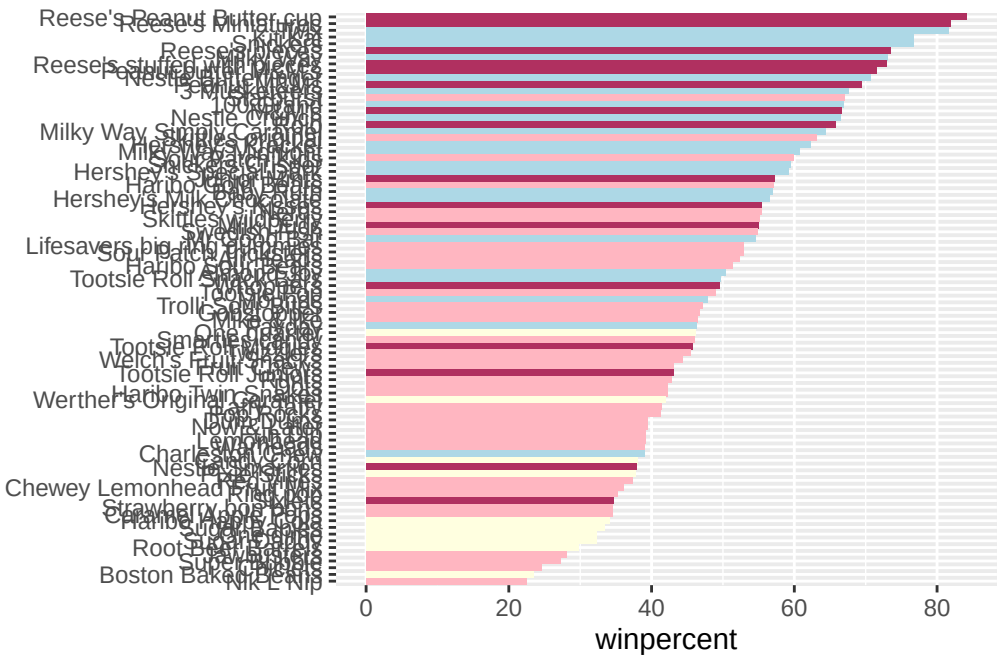
I want custom colours that I pick so we need to make this ourselves

```r
my_cols <- rep("lightyellow", nrow(candy))
my_cols[candy$chocolate==1] <- "maroon"
my_cols[candy$fruity==1] <- "lightpink"
my_cols[candy$bar==1] <- "lightblue"
my_cols
```

```
 [1] "lightblue"   "lightblue"   "lightyellow" "lightyellow" "lightpink"
 [6] "lightblue"   "lightblue"   "lightyellow" "lightyellow" "lightpink"
[11] "lightblue"   "lightpink"   "lightpink"   "lightpink"   "lightpink"
[16] "lightpink"   "lightpink"   "lightpink"   "lightpink"   "lightyellow"
[21] "lightpink"   "lightpink"   "maroon"      "lightblue"   "lightblue"
[26] "lightblue"   "lightpink"   "maroon"      "lightblue"   "lightpink"
[31] "lightpink"   "lightpink"   "maroon"      "maroon"      "lightpink"
[36] "maroon"      "lightblue"   "lightblue"   "lightblue"   "lightblue"
```

```
[41] "lightblue"   "lightpink"   "lightblue"   "lightblue"   "lightpink"
[46] "lightpink"   "lightblue"   "maroon"      "lightyellow" "lightpink"
[51] "lightpink"   "maroon"      "maroon"      "maroon"      "maroon"
[56] "lightpink"   "maroon"      "lightyellow" "lightpink"   "maroon"
[61] "lightpink"   "lightpink"   "maroon"      "lightpink"   "lightblue"
[66] "lightblue"   "lightpink"   "lightpink"   "lightpink"   "lightpink"
[71] "lightyellow" "lightyellow" "lightpink"   "lightpink"   "lightpink"
[76] "maroon"      "maroon"      "lightblue"   "lightpink"   "lightblue"
[81] "lightpink"   "lightpink"   "lightpink"   "lightyellow" "maroon"
```

```
ggplot(candy) + aes(winpercent,
                    reorder( rownames(candy), winpercent),
                     ) +
  geom_col(fill=my_cols) +
  ylab("") # turn y label off that we dont need
```



Q17. What is the worst ranked chocolate candy?

Sixlets

Q18. What is the best ranked fruity candy?

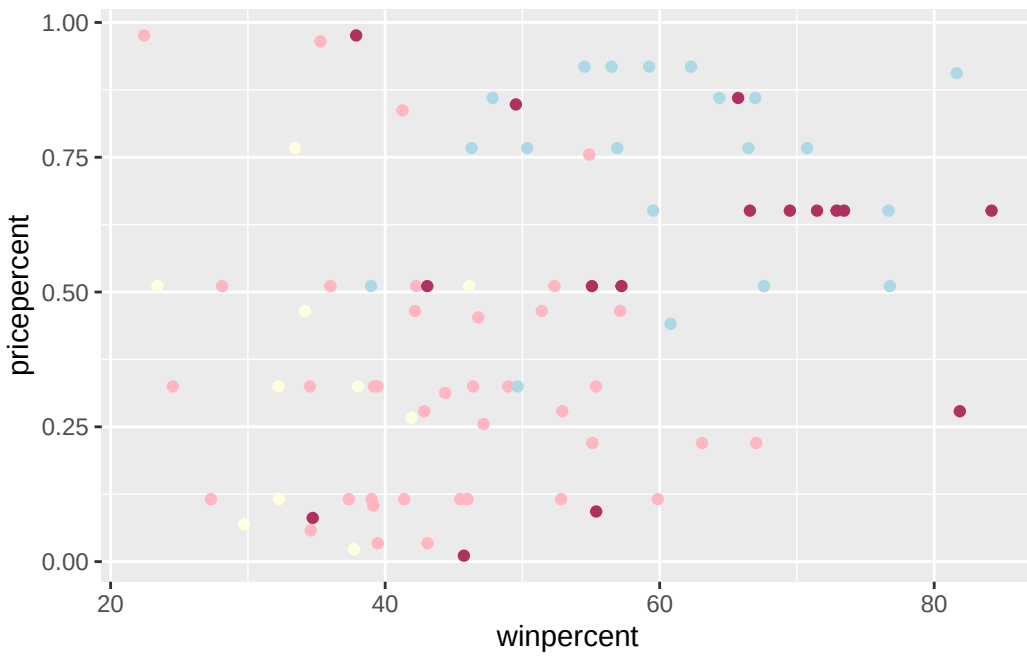
Starbursts

Taking a Look at Pricepercent

What about value for money? What is the best candy for the least money? One way to get at this would be to make a plot of winpercent vs the pricepercent variable. The pricepercent variable records the percentile rank of the candy's price against all the other candies in the dataset. Lower values are less expensive and higher values are more expensive.

Make a plot of winpercent vs pricepercent

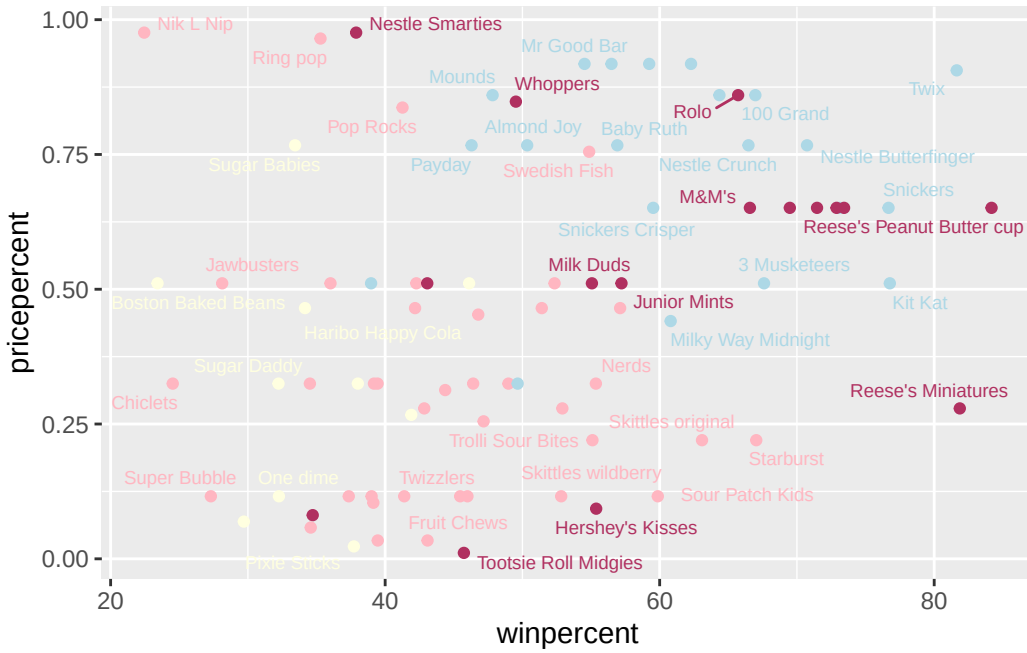
```
ggplot(candy) +
  aes(winpercent, pricepercent) +
  geom_point(col=my_cols)
```



We can use the **ggrepel** package for better label placement

```
library(ggrepel)
ggplot(candy) +
  aes(winpercent, pricepercent, label=rownames(candy)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols, size=2.5, max.overlaps=8)
```

```
Warning: ggrepel: 40 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



> Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's miniatures

> Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

Nik L Nip- least popular and very expensive Ring pop, Nestle smarties, Pop Rocks and Sugar Babies

Exploring the correlation structure

Pearson correlation values range from -1 to +1

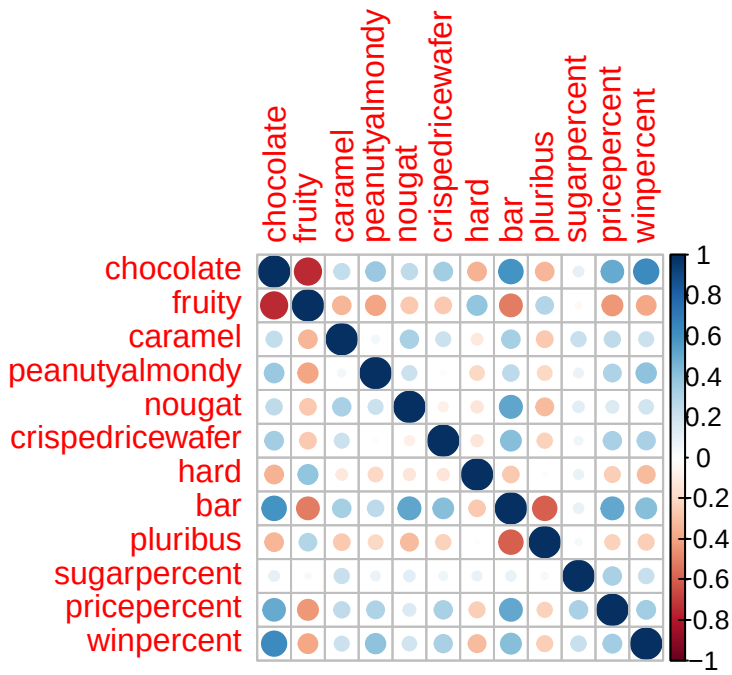
```
library(corrplot)
```

```
corrplot 0.95 loaded
```

```
cor(candy)
```

```
                  chocolate      fruity     caramel peanutyalmondy      nougat
chocolate         1.0000000 -0.74172106  0.24987535     0.37782357  0.25489183
fruity           -0.7417211  1.00000000 -0.33548538    -0.39928014 -0.26936712
caramel           0.2498753 -0.33548538  1.00000000     0.05935614  0.32849280
peanutyalmondy    0.3778236 -0.39928014  0.05935614     1.00000000  0.21311310
nougat            0.2548918 -0.26936712  0.32849280     0.21311310  1.00000000
crispedricewafer  0.3412098 -0.26936712  0.21311310    -0.01764631 -0.08974359
hard             -0.3441769  0.39067750 -0.12235513    -0.20555661 -0.13867505
bar               0.5974211 -0.51506558  0.33396002     0.26041960  0.52297636
pluribus         -0.3396752  0.29972522 -0.26958501    -0.20610932 -0.31033884
sugarpercent      0.1041691 -0.03439296  0.22193335     0.08788927  0.12308135
pricepercent      0.5046754 -0.43096853  0.25432709     0.30915323  0.15319643
winpercent        0.6365167 -0.38093814  0.21341630     0.40619220  0.19937530
                 crispedricewafer        hard         bar    pluribus
chocolate              0.34120978 -0.34417691  0.59742114 -0.33967519
fruity                -0.26936712  0.39067750 -0.51506558  0.29972522
caramel                0.21311310 -0.12235513  0.33396002 -0.26958501
peanutyalmondy        -0.01764631 -0.20555661  0.26041960 -0.20610932
nougat                -0.08974359 -0.13867505  0.52297636 -0.31033884
crispedricewafer       1.00000000 -0.13867505  0.42375093 -0.22469338
hard                  -0.13867505  1.00000000 -0.26516504  0.01453172
bar                    0.42375093 -0.26516504  1.00000000 -0.59340892
pluribus              -0.22469338  0.01453172 -0.59340892  1.00000000
sugarpercent           0.06994969  0.09180975  0.09998516  0.04552282
pricepercent           0.32826539 -0.24436534  0.51840654 -0.22079363
winpercent             0.32467965 -0.31038158  0.42992933 -0.24744787
                 sugarpercent pricepercent winpercent
chocolate          0.10416906    0.5046754  0.6365167
fruity            -0.03439296   -0.4309685 -0.3809381
caramel            0.22193335    0.2543271  0.2134163
peanutyalmondy     0.08788927    0.3091532  0.4061922
nougat             0.12308135    0.1531964  0.1993753
crispedricewafer   0.06994969    0.3282654  0.3246797
hard               0.09180975   -0.2443653 -0.3103816
bar                0.09998516    0.5184065  0.4299293
pluribus           0.04552282   -0.2207936 -0.2474479
sugarpercent       1.00000000    0.3297064  0.2291507
pricepercent       0.32970639    1.0000000  0.3453254
winpercent         0.22915066    0.3453254  1.0000000
```

```r
cij <- cor(candy)
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

chocolate and fruity, and bar and pluribus

Q23. Similarly, what two variables are most positively correlated?

chocolate bar and winpercent chocolate and bar, bar and peanutyalmondy

Principal Componenet Analysis

REMEMBER SCALE=TRUE!

```
pca <- prcomp(candy, scale=TRUE)
```

(candy is input)

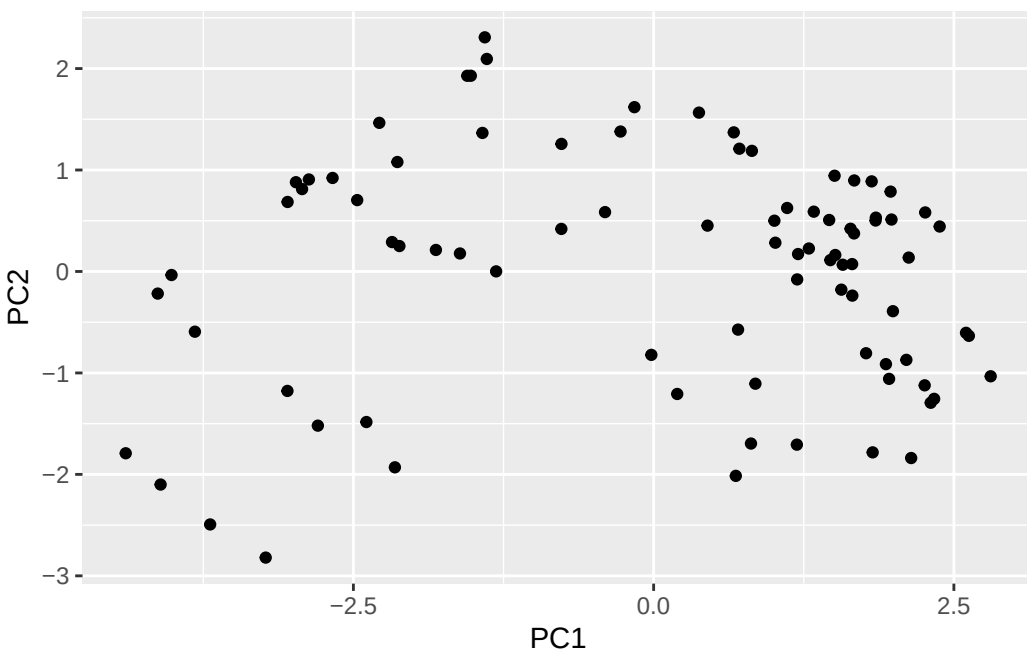
```
summary(pca)
```

```
Importance of components:
                          PC1    PC2    PC3     PC4    PC5     PC6     PC7
Standard deviation     2.0788 1.1378 1.1092 1.07533 0.9518 0.81923 0.81530
```

```
Proportion of Variance 0.3601 0.1079 0.1025 0.09636 0.0755 0.05593 0.05539
Cumulative Proportion  0.3601 0.4680 0.5705 0.66688 0.7424 0.79830 0.85369
                           PC8     PC9    PC10    PC11    PC12
Standard deviation     0.74530 0.67824 0.62349 0.43974 0.39760
Proportion of Variance 0.04629 0.03833 0.03239 0.01611 0.01317
Cumulative Proportion  0.89998 0.93832 0.97071 0.98683 1.00000
```

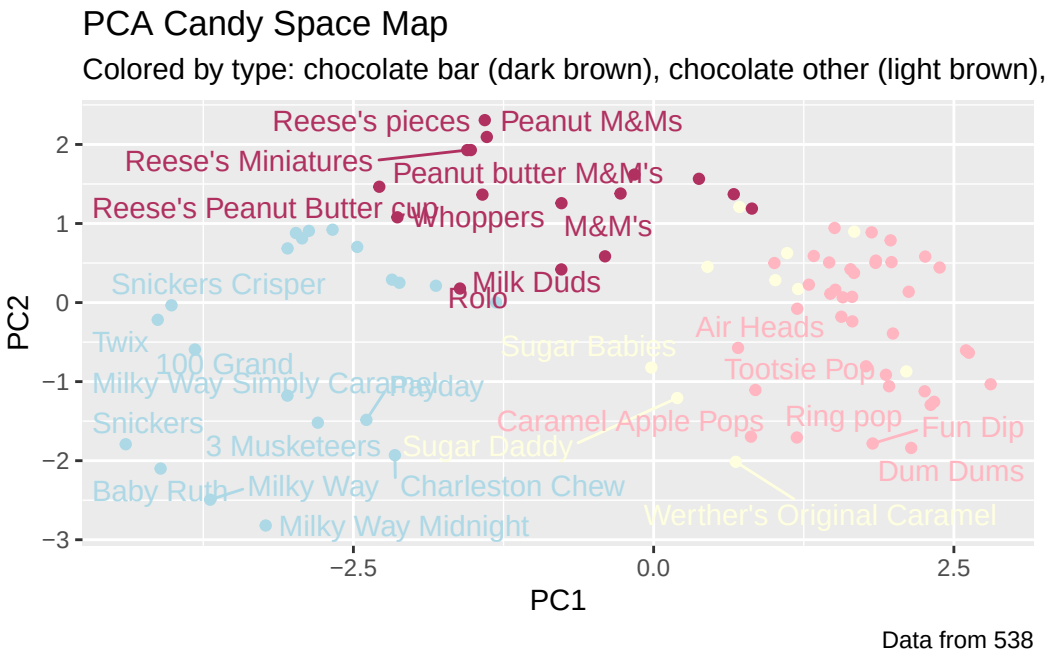
The main results figure: the PCA score plot:

```
ggplot(pca$x) +
  aes(PC1, PC2) +
  geom_point()
```



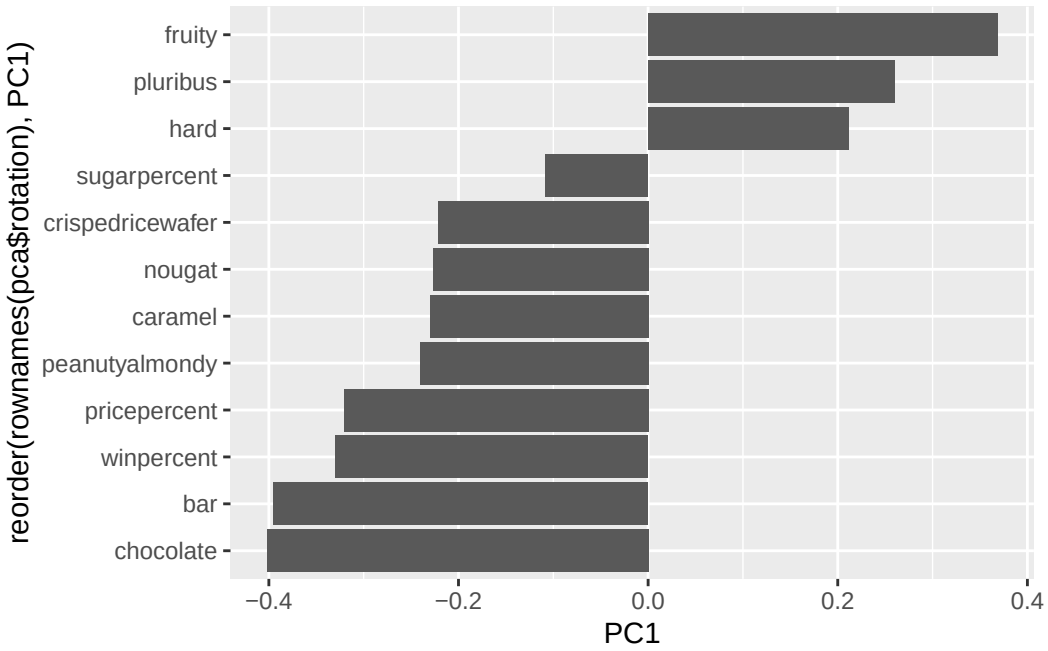

```
ggplot(pca$x) +
  aes(PC1, PC2, label=rownames(pca$x)) +
  geom_point(col=my_cols) +
  geom_text_repel(col=my_cols) +
  labs(title="PCA Candy Space Map",
       subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),
       caption="Data from 538")
```

```
Warning: ggrepel: 56 unlabeled data points (too many overlaps). Consider
increasing max.overlaps
```



The "loadings" plot for PC1

```
ggplot(pca$rotation) +
  aes(PC1, reorder (rownames(pca$rotation), PC1)) +
  geom_col()
```



> Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

the fruity, hard, pluribus fruits. Yes these make sense, as in our corrplot, they had correlation too.

> Q25. Based on your exploratory analysis, correlation findings, and PCA results, what combination of characteristics appears to make a "winning" candy? How do these different analyses (visualization, correlation, PCA) support or complement each other in reaching this conclusion?

Through our analysis, chocolate are the 'winning' candy. We can see this through the two means of `winpercent` of choc and fruit, with chocolate having a higher mean. the corrplot showed that chocolate has a higher correlation with winpercent than fruit. chocolate bars seem to have the highest distribution in the higher value of winpercent.